

Fairness

Machine Learning for Behavioral Data

April 28, 2026

Today's Topic

Week	Lecture/Lab
9	Time Series Clustering
10	Explainability
11	Fairness
12	No Lecture / Team Discussions
13	No Lecture / Office Hours
14	No Lecture / Office Hours
15	Poster Session

Getting ready for today's lecture...

➤ Join us on SpeakUp

<https://go.epfl.ch/mlbd-speakup>



Agenda

- 1) **Introduction to ethics in ML**
- 2) Fairness in machine learning:
 - Sources of unfairness
 - Fairness metrics – evaluating model predictions
- 3) Example on real world data (if time)

Fairness in ML game

EPFL has developed a game-based approach to introduce students to the topic of fairness in ML

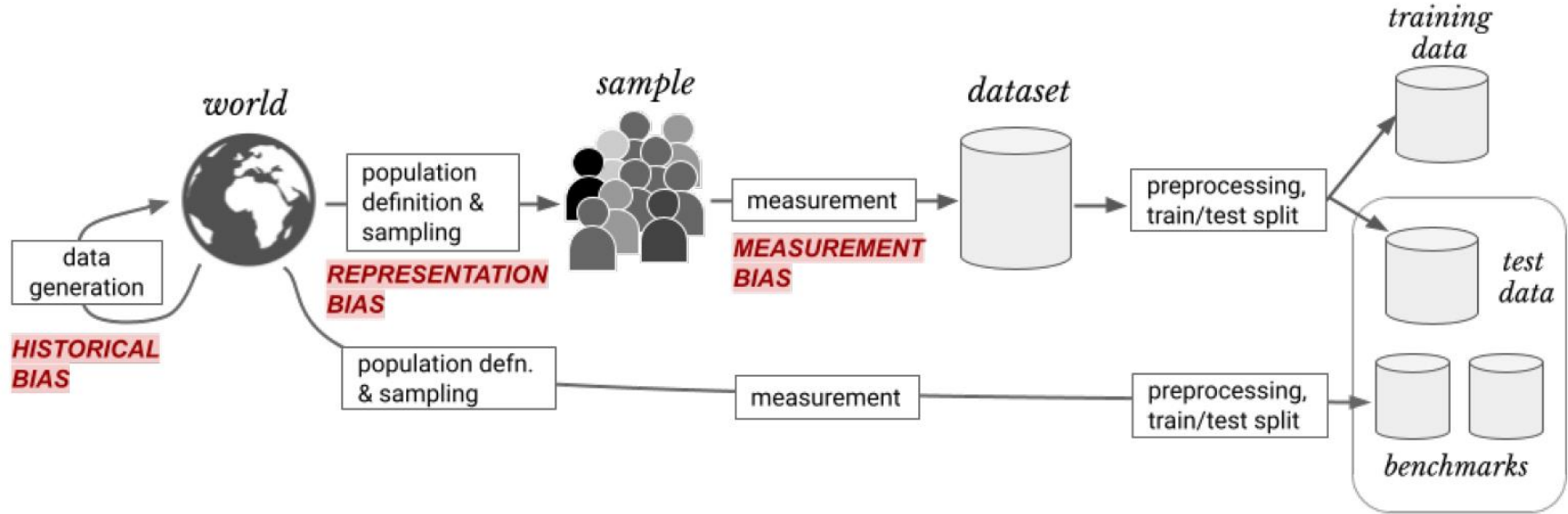
<https://companion.epfl.ch/game/7d7zs7/>



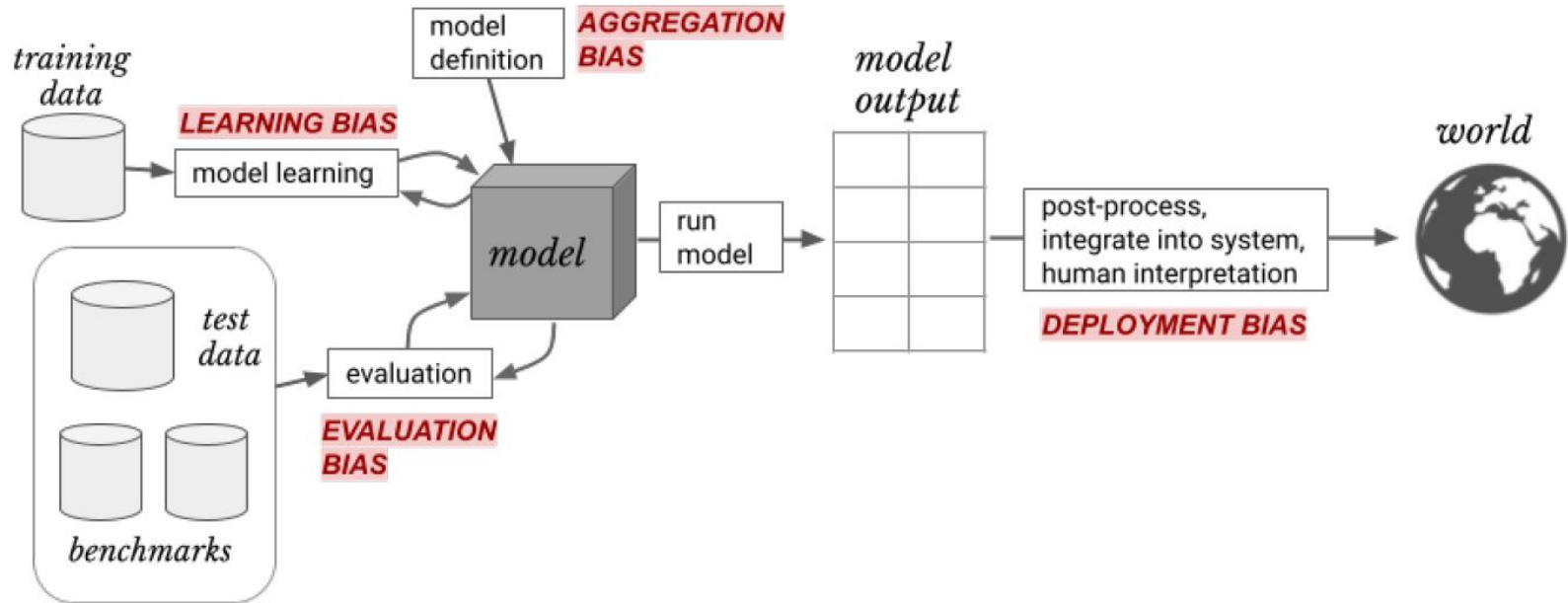
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Sources of Unfairness – Data Generation

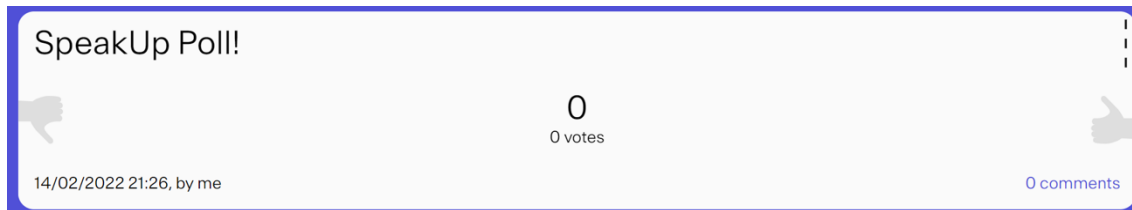


Sources of Unfairness – Model Building



Fairness Through Blindness

- Idea: we ignore all protected attributes in our model (e.g., we do not use protected attributes such as gender, race, etc. as features)



Will this idea lead to a fair model?

a) Yes

b) No

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Fairness Through Awareness

- There is not one mathematically agreed definition of fairness
- Popular fairness metrics are
 - model-agnostic
 - defined for classification problems

Problem Formalization

Notation:

- X is the input to the model
- $\hat{Y}$ is the prediction of the model
- T is the true label
- A is the protected attribute

Confusion Matrix

		True Label	
		$T = 1$	$T = 0$
Predicted Label	$Y = 1$	True Positive (TP)	False Positive (FP)
	$Y = 0$	False Negative (FN)	True Negative (TN)

Demographic Parity

- Requires equal proportion of positive predictions in each group

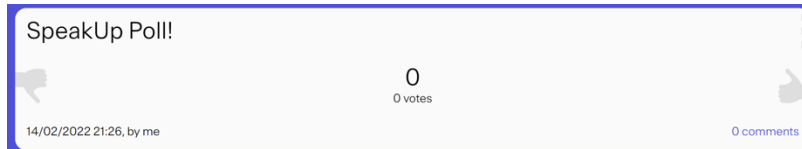
$$p(\hat{Y} = 1 | A = 1) = p(\hat{Y} = 1 | A = 0)$$

Demographic Parity - Example

- Admittance to *Fruits* University
- Students from two schools apply: *Apple* and *Peach*
- School *Peach* has a better program, resulting in more qualified students

<i>Peach</i>	Qualified	Unqualified
Admitted	45	2
Rejected	45	8

<i>Apple</i>	Qualified	Unqualified
Admitted	5	18
Rejected	5	72



Is demographic parity fulfilled?

a) Yes

b) No

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Equalized Odds

- For any label and attribute, a classifier predicts the label equally well for all values of that attribute

$$p(\hat{Y} = 1 | A = 1, T = 1) = p(\hat{Y} = 1 | A = 0, T = 1)$$

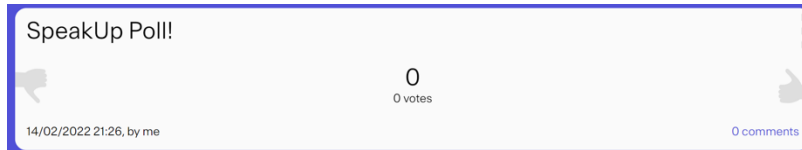
$$p(\hat{Y} = 1 | A = 1, T = 0) = p(\hat{Y} = 1 | A = 0, T = 0)$$

Equalized Odds - Example

- Admittance to *Fruits* University
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Are equalized odds fulfilled?

a) Yes

b) No

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Predictive Value Parity

- Probability of a sample with positive (negative) predictive value to truly belong to the positive (negative) class should be the same across attributes

$$p(T = 1|A = 1, \hat{Y} = 1) = p(T = 1|A = 0, \hat{Y} = 1)$$

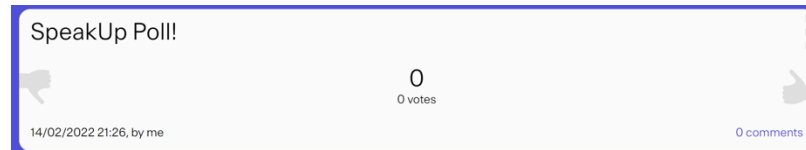
$$p(T = 0|A = 1, \hat{Y} = 0) = p(T = 0|A = 0, \hat{Y} = 0)$$

Predictive Value Parity - Example

- Admittance to *Fruits* University
- Students from two schools apply: *Apple* and *Peach*
- School *Peach* has a better program, resulting in more qualified students

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Is predictive value parity fulfilled?

a) Yes

b) No

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Impossibility Result

- Any two of the three criteria are mutually exclusive
 1. If A and T are not independent, then *demographic parity* and *predictive value* parity cannot simultaneously hold
 2. If A and $\hat{Y}$ are not independent of T , then *demographic parity* and *equalized odds* cannot simultaneously hold
 3. If A and T are not independent, then *equalized odds* and *predictive value parity* cannot simultaneously hold

Impossibility Result

Note that these requirements hold for *most* classifiers in real contexts:

- Base-rates of outcomes rarely are equal across groups
- A and T are usually associated when issues of fairness are relevant for the group in question
- $\hat{Y}$ and T are usually associated, if your classifier is any good

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Flipped Classroom – Your Turn

- Participants: 214 EPFL students of a course taught in *flipped classroom* mode with a duration of 10 weeks
- We have trained a classifier to predict whether a student will pass or fail the course based on their clickstream data
- Your task:
 1. Choose one of the fairness metrics introduced in class and compute the metric for the flipped classroom classifier
 2. Tell us: is the classifier fair according to the selected metric? Why did you choose this metric?

Summary

- There are multiple sources of unfairness in a machine learning pipeline
- There is no consensus on the mathematical definition of fairness metrics
- Different metrics assess different aspects of the classifier
- Often, fairness metrics are mutually exclusive
- Fairness evaluation of a classifier includes exploration of relevant characteristics of our data